

ENME608 Mechanisms

Acceleration Analysis (Norton Ch. 7)

Aaron Taylor — 24232594

2026

Contents

1	Acceleration Analysis — Overview	2
2	Four-Bar Angular Accelerations	2
3	Acceleration of Points	2
4	Slider-Crank Accelerations	2
5	Coriolis Acceleration	3
6	Tutorial Solutions (Tutorial 6 — Acceleration)	3
6.1	P7-4 — Four-Bar Accelerations (Row a)	3
6.2	P7-6 — Offset Crank-Slider Accelerations (Row a)	4

Acceleration Analysis — Overview

Acceleration analysis is obtained by *differentiating the vector loop equation twice*. Each link contributes two acceleration components:

- **Tangential** $a^t = R\alpha$, perpendicular to the link (direction $(-\sin \theta, \cos \theta)$);
- **Normal (centripetal)** $a^n = R\omega^2$, pointing from the point back toward the centre of rotation (direction $(\cos \theta, \sin \theta)$).

Differentiating $Re^{j\theta}$ twice gives $R(j\alpha - \omega^2)e^{j\theta}$ — the $j\alpha$ term is tangential, the $-\omega^2$ term is normal.

Prerequisite: position (θ_3, θ_4) and velocity (ω_3, ω_4) results, same assembly mode.

Open-book: Lec 5 (acceleration), Tutorial 6.

Exam focus. Always carry the **normal** terms — forgetting $R\omega^2$ is the most common error. Even when $\alpha_2 = 0$ (constant input speed), the normal accelerations are non-zero and drive the whole solution.

Four-Bar Angular Accelerations

Differentiating the four-bar velocity loop once more:

$$a(j\alpha_2 - \omega_2^2)e^{j\theta_2} + b(j\alpha_3 - \omega_3^2)e^{j\theta_3} - c(j\alpha_4 - \omega_4^2)e^{j\theta_4} = 0.$$

Separating into real and imaginary parts leaves two linear equations in the unknowns α_3, α_4 . Norton's solution uses the coefficients

$$\begin{aligned} A &= c \sin \theta_4, & B &= b \sin \theta_3, \\ C &= a\alpha_2 \sin \theta_2 + a\omega_2^2 \cos \theta_2 + b\omega_3^2 \cos \theta_3 - c\omega_4^2 \cos \theta_4, \\ D &= c \cos \theta_4, & E &= b \cos \theta_3, \\ F &= a\alpha_2 \cos \theta_2 - a\omega_2^2 \sin \theta_2 - b\omega_3^2 \sin \theta_3 + c\omega_4^2 \sin \theta_4, \\ \alpha_3 &= \frac{CD - AF}{AE - BD}, & \alpha_4 &= \frac{CE - BF}{AE - BD}. \end{aligned}$$

Acceleration of Points

Pin A:

$$\mathbf{a}_A = a\alpha_2(-\sin \theta_2, \cos \theta_2) - a\omega_2^2(\cos \theta_2, \sin \theta_2).$$

Pin B (difference equation):

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{BA}, \quad \mathbf{a}_{BA} = b\alpha_3(-\sin \theta_3, \cos \theta_3) - b\omega_3^2(\cos \theta_3, \sin \theta_3).$$

Coupler point P (R_{PA} at orientation $\theta_3 + \delta_3$):

$$\mathbf{a}_P = \mathbf{a}_A + R_{PA}\alpha_3(-\sin(\theta_3 + \delta_3), \cos(\theta_3 + \delta_3)) - R_{PA}\omega_3^2(\cos(\theta_3 + \delta_3), \sin(\theta_3 + \delta_3)).$$

Slider-Crank Accelerations

For the offset slider-crank (constant offset, slider along X):

$$\alpha_3 = \frac{a\omega_2^2 \sin \theta_2 - a\alpha_2 \cos \theta_2 + b\omega_3^2 \sin \theta_3}{b \cos \theta_3},$$

$$\ddot{d} = -a\alpha_2 \sin \theta_2 - a\omega_2^2 \cos \theta_2 - b\alpha_3 \sin \theta_3 - b\omega_3^2 \cos \theta_3.$$

$\ddot{d}$ is the slider (and slip) acceleration along the slide axis.

Coriolis Acceleration

When a point *slides along a rotating link*, the total acceleration of the moving point relative to ground contains **four** terms (Norton’s full relative-acceleration equation):

$$\mathbf{a}_P = \underbrace{\mathbf{a}_{P'}}_{\text{coincident pt}} + \underbrace{\mathbf{a}_{\text{rel}}}_{\text{slip along link}} + \underbrace{2\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}}}_{\text{Coriolis}} + \underbrace{(\text{normal/tangential of transport})}_{\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}), \boldsymbol{\alpha} \times \mathbf{r}}.$$

The **Coriolis component** has magnitude

$$a^{\text{cor}} = 2\omega v_{\text{rel}},$$

directed by rotating the slip-velocity vector 90° in the direction of ω . It appears *only* when there is simultaneous rotation of the link *and* sliding along it (e.g. inverted slider-crank, Geneva mechanism, quick-return Whitworth).

Exam focus. Coriolis is easy to miss because it has no analogue in pure-pin linkages. Trigger: “a slider moving on a member that is itself rotating.” If $\omega = 0$ or $v_{\text{rel}} = 0$, the Coriolis term vanishes.

Tutorial Solutions (Tutorial 6 — Acceleration)

P7-4 — Four-Bar Accelerations (Row a)

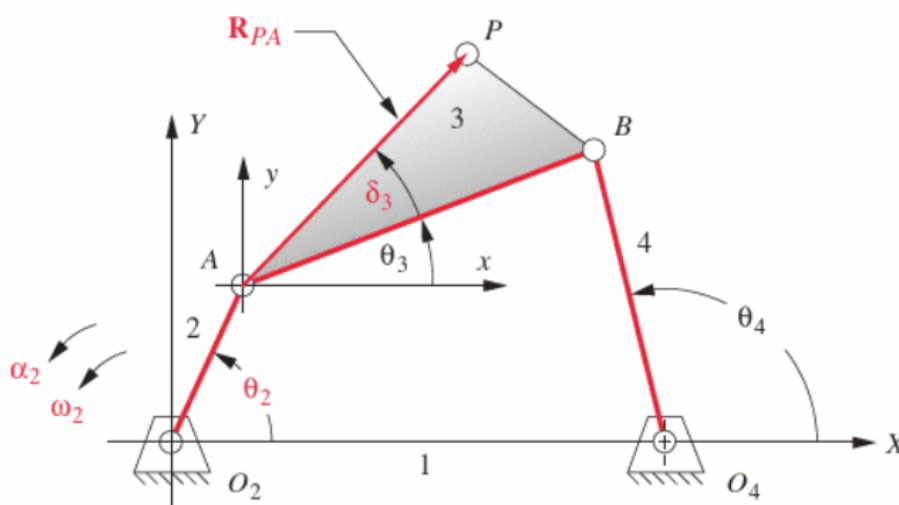


FIGURE P7-1

Figure P7-1: general four-bar with input α_2, ω_2 and coupler point P .

Given:

Row a (Table P7-1): $a = L_2 = 2$, $b = L_3 = 7$, $c = L_4 = 9$, $d = L_1 = 6$, $\theta_2 = 30^\circ$, $\omega_2 = 10$ rad/s, $\alpha_2 = 0$, $R_{PA} = 6$, $\delta_3 = 30^\circ$. From position/velocity: $\theta_3 = 88.8^\circ$, $\theta_4 = 117.3^\circ$, $\omega_3 = -5.99$, $\omega_4 = -3.99$ rad/s.

Find:

α_3 , α_4 , and the accelerations of A , B , P (open configuration).

Solution:

Coefficients (with $\alpha_2 = 0$):

$$A = 9 \sin 117.3^\circ = 8.00, \quad B = 7 \sin 88.8^\circ = 7.00, \quad D = 9 \cos 117.3^\circ = -4.13, \quad E = 7 \cos 88.8^\circ = 0.142,$$

$$C = a\omega_2^2 \cos \theta_2 + b\omega_3^2 \cos \theta_3 - c\omega_4^2 \cos \theta_4 = 173.2 + 5.1 + 65.7 = 244.0,$$

$$F = -a\omega_2^2 \sin \theta_2 - b\omega_3^2 \sin \theta_3 + c\omega_4^2 \sin \theta_4 = -100 - 251.1 + 127.3 = -223.8.$$

With $AE - BD = (8.00)(0.142) - (7.00)(-4.13) = 30.0$:

$$\alpha_3 = \frac{CD - AF}{AE - BD} = \frac{(244.0)(-4.13) - (8.00)(-223.8)}{30.0} = \boxed{26.1 \text{ rad/s}^2},$$

$$\alpha_4 = \frac{CE - BF}{AE - BD} = \frac{(244.0)(0.142) - (7.00)(-223.8)}{30.0} = \boxed{53.3 \text{ rad/s}^2}.$$

Point accelerations ($\alpha_2 = 0$, so $\mathbf{a}_A$ is purely centripetal):

$$\mathbf{a}_A = -(2)(10^2)(\cos 30^\circ, \sin 30^\circ) = (-173.2, -100) \text{ in/s}^2, \quad |\mathbf{a}_A| = 200.$$

$$\mathbf{a}_{BA} = (7)(26.1)(-\sin 88.8^\circ, \cos 88.8^\circ) - (7)(5.99^2)(\cos 88.8^\circ, \sin 88.8^\circ) = (-187.5, -247.4),$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{BA} = (-360.7, -347.4) \text{ in/s}^2, \quad \boxed{|\mathbf{a}_B| = 501 \text{ in/s}^2}.$$

Check from link 4: $\mathbf{a}_B = (9)(53.3)(-\sin 117.3^\circ, \cos 117.3^\circ) - (9)(3.99^2)(\cos 117.3^\circ, \sin 117.3^\circ) = (-360.7, -347.4) \checkmark$

$$\mathbf{a}_P = \mathbf{a}_A + (6)(26.1)(-\sin 118.8^\circ, \cos 118.8^\circ) - (6)(5.99^2)(\cos 118.8^\circ, \sin 118.8^\circ) = (-206, -364),$$

$$\boxed{|\mathbf{a}_P| = 418 \text{ in/s}^2 \text{ at } 240.5^\circ}.$$

P7-6 — Offset Crank-Slider Accelerations (Row a)

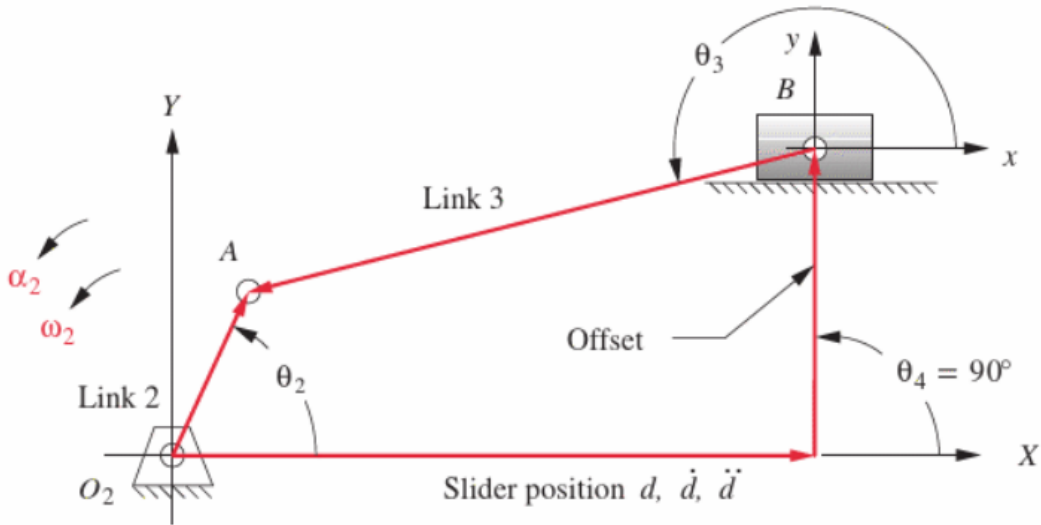


FIGURE P7-2

Figure P7-2: noninverted offset crank-slider; slider position $d, \dot{d}, \ddot{d}$.

Given:

Row a (Table P7-2): crank $a = L_2 = 1.4$, coupler $b = L_3 = 4$, offset $= 1$, $\theta_2 = 45^\circ$, $\omega_2 = 10 \text{ rad/s}$, $\alpha_2 = 0$. From position/velocity: $\theta_3 = 0.14^\circ$, $\omega_3 = -2.48 \text{ rad/s}$.

Find:

α_3 , accelerations of pins A and B, and the acceleration of slip.

Solution:

$$\alpha_3 = \frac{a\omega_2^2 \sin \theta_2 + b\omega_3^2 \sin \theta_3}{b \cos \theta_3} = \frac{(1.4)(100) \sin 45^\circ + (4)(2.48^2) \sin 0.14^\circ}{4 \cos 0.14^\circ} = \boxed{24.8 \text{ rad/s}^2}.$$

$$\mathbf{a}_A : \quad |\mathbf{a}_A| = a\omega_2^2 = (1.4)(100) = 140 \text{ in/s}^2 \text{ (toward } O_2).$$

$$\ddot{d} = -a\omega_2^2 \cos \theta_2 - b\alpha_3 \sin \theta_3 - b\omega_3^2 \cos \theta_3 = -98.99 - 0.06 - 24.6 = \boxed{-123.7 \text{ in/s}^2}.$$

The slider B translates only, so $\mathbf{a}_B = \ddot{d}$ along the slide and the **acceleration of slip** at the sliding joint is $\boxed{123.7 \text{ in/s}^2}$ ($-X$ direction). (No Coriolis term here: the slide axis does not rotate.)